

课程笔记 SNN训练算法汇总综述

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摘要

课程笔记 SNN训练算法汇总综述，涵盖了从传统的 BPTT/STBP 到最新的在线学习方法（OTTT、NDOT）以及局部学习规则（S-TLLR、TESS）。重点分析了每种方法的核心思想、数学推导、优缺点以及适用场景。通过对比不同算法在时间依赖性处理、梯度计算复杂度和生物可解释性方面的差异，全面总结了 SNN 训练领域的研究进展和未来方向。

PPT Week4_SNN Training.pptx

文章汇总

RTRL: A learning algorithm for continually running fully recurrent neural networks [1];

E-prop: A solution to the learning dilemma for recurrent networks of spiking neurons [2];

OTTT: Online Training Through Time for Spiking Neural Networks [3];

NDOT: NDOT: Neuronal Dynamics-based Online Training for Spiking Neural Networks [4];

S-TLLR: S-TLLR: STDP-inspired Temporal Local Learning Rule for Spiking Neural Networks [5];

TESS: TESS: A Scalable Temporally and Spatially Local Learning Rule for Spiking Neural Networks [6]

I. ELIGIBILITY TRACE: REPLACING BPTT MEMORY

Eligibility Trace: Replacing BPTT Memory

The Bottleneck of BPTT:

- Unrolls the computational graph over T time steps.
- Memory complexity: $\mathcal{O}(T \times n)$ (stores all states).
- Prevents online / on-device learning.

Core Idea of Eligibility Trace:

- Replace historical memory with a *recursive state*.
- Accumulates past effects into a local "trace" e_{ji}^t .
- Enables *forward-in-time* gradient computation.

Unified Mathematical Form:

$$\Delta W_{ji} = \sum_t L_j^t \cdot e_{ji}^t \quad (1)$$

- L_j^t : Top-down learning signal (error/reward).

- e_{ji}^t : Eligibility trace (local synaptic history).

讲稿对应本页PPT:

”本页介绍在线学习算法的核心动机——替代 BPTT 的内存瓶颈。BPTT 需要展开所有时间步并存储中间状态，内存为 $O(Tn)$ 。Eligibility Trace 的核心思想是用一个递归状态替代完整历史记忆。统一数学形式为 $\Delta W = \sum L \cdot e$ ，其中 L 是顶向下学习信号， e 是局部资格迹。”

II. REPRESENT WORK—RTRL: REAL-TIME RECURRENT LEARNING

RTRL: Real-Time Recurrent Learning (1989)

Key Innovation: Computes *exact* gradients online (no unrolling).

Gradient Sensitivity Dynamics:

$$p_{ij}^k(t+1) = f'_k(s_k(t)) \left[\sum_{l \in U} w_{kl} p_{ij}^l(t) + \delta_{ik} z_j(t) \right] \quad (2)$$

- $p_{ij}^k(t) \triangleq \frac{\partial y_k(t)}{\partial w_{ij}}$: Sensitivity.
- $\sum_l w_{kl} p_{ij}^l(t)$: **Recursive eligibility** (past influences).
- $\delta_{ik} z_j(t)$: Immediate presynaptic input.

Weight Update:

$$\Delta w_{ij}(t) = \alpha \sum_{k \in U} e_k(t) p_{ij}^k(t) \quad (3)$$

Limitation: Memory Complexity: $\mathcal{O}(n^3)$ (impractical for large nets).

讲稿对应本页PPT:

”RTRL 是 Williams 和 Zipser 于 1989 年提出的开山之作，核心创新是首次实现了精确梯度的在线计算。公式中 p_{ij}^k 表示输出对权重的敏感度。其递归项 $\sum w \cdot p$ 将过去影响向前传递，构成资格迹的雏形。权重的更新为误差 e_k 乘以敏感度 p 。但其致命局限是内存复杂度高达 $\mathcal{O}(n^3)$ ，无法用于大规模网络。”

III. REPRESENT WORK—E-PROP

e-Prop: Eligibility Propagation (2020)

Key Innovation: Biologically plausible approximation using neuron-specific feedback.

Fundamental Factorization:

$$\frac{dE}{dW_{ji}} = \sum_t L_j^t \cdot e_{ji}^t \quad (4)$$

- $L_j^t \triangleq \frac{\partial E}{\partial z_j^t}$: Learning signal.
- $e_{ji}^t \triangleq \left[\frac{dz_j^t}{dW_{ji}} \right]_{\text{local}}$: Eligibility trace.

Eligibility Trace for ALIF (Adaptive) Neurons:

$$e_{ji}^t = \psi_j^t (\bar{z}_i^{t-1} - \beta \epsilon_{ji,a}^t) \quad (5)$$

- ψ_j^t : Surrogate gradient (pseudo-derivative).
- $\epsilon_{ji,a}^t$: Slow component from threshold adaptation.

Key Insight: Slow hidden variables provide “highways into the future” for gradients.

讲稿对应本页PPT:

”e-Prop 由 Bellec 等人于 2020 年提出，核心是提供了生物合理的 BPTT 近似。其基础分解式为 $dE/dW = \sum L \cdot e$ 。针对具有自适应阈值的 ALIF 神经元，其资格迹 e 包含代理梯度 ψ 和慢变量 ϵ 。这个慢变量来自阈值适应，能将梯度信息在长时间尺度上向前传播，充当了通往未来的‘高速公路’，这是 e-Prop 在生物合理性上的关键洞察。”

IV. REPRESENT WORK—OTTT (ONLINE TRAINING THROUGH TIME)

OTTT: Online Training Through Time (2022)

Key Innovation: Decouples temporal dependency to achieve $\mathcal{O}(n)$ memory.

Decoupled Gradient Formulation:

$$\nabla_{\mathbf{W}^l} L[t] = \mathbf{g}_{\mathbf{u}^{l+1}}[t] \cdot (\hat{\mathbf{a}}^l[t])^\top \quad (6)$$

- $\mathbf{g}_{\mathbf{u}^{l+1}}[t]$: Instantaneous spatial gradient.
- $\hat{\mathbf{a}}^l[t]$: Presynaptic activity trace (temporal gradient).

Recursive Trace Update:

$$\hat{a}^l[t+1] = \lambda \hat{a}^l[t] + s^l[t+1] \quad (7)$$

- Tracks exponentially decaying sum of presynaptic spikes.
- Replaces expensive $\prod \frac{\partial u}{\partial u}$ terms.

Theoretical Guarantee:

- OTTT gradients provide a valid *descent direction* for optimization.

讲稿对应本页PPT:

”OTTT 由 Xiao 等人于 2022 年提出，核心创新是将时间依赖解耦，实现 $\mathcal{O}(n)$ 常数内存。其解耦梯度为空间梯度 g 乘以时间梯度 $\hat{a}$ 。 $\hat{a}$ 的递归更新为 $\hat{a}[t+1] = \lambda \hat{a}[t] + s[t+1]$ ，即指数衰减的突触前活动之和，替代了 BPTT 中复杂的连乘项。此外，OTTT 从理论上证明了其梯度能提供有效的下降方向。”

V. REPRESENT WORK—NDOT (NEURONAL DYNAMICS-BASED ONLINE TRAINING)

NDOT: Neuronal Dynamics-based Online Training (2024)

Key Innovation: Uses *continuous* neuronal dynamics for precise temporal dependency.

Continuous Temporal Dependency (Derivative Ratio):

$$e^l[t] \triangleq \frac{\partial \mathbf{u}^l[t+1]}{\partial \mathbf{u}^l[t]} = \frac{\mathbf{u}^l[t] - V_{th} \mathbf{s}^l[t]}{\mathbf{u}^l[t-1] - V_{th} \mathbf{s}^l[t-1]} \quad (8)$$

- Derived from LIF dynamics: $e(t) = u'(t+1) \oslash u'(t)$.
- $\oslash$ denotes element-wise division.

Precise Temporal Gradient Tracking:

$$\hat{\mathbf{a}}^{l-1}[t] = \mathbf{e}^{l-1}[t-1] \odot \hat{\mathbf{a}}^{l-1}[t-1] + \mathbf{s}^{l-1}[t] \quad (9)$$

- Uses dynamic $\mathbf{e}^{l-1}[t-1]$ (vs fixed λ in OTTT).
- Achieves higher accuracy via exact history modeling.

讲稿对应本页PPT:

”NDOT 由 Jiang 等人于 2024 年提出，核心是使用神经元动力学的连续形式计算精确的时间依赖。公式中 $e^l[t]$ 定义为膜电位对时间的导数之比，源自 LIF 连续动力学。与传统 OTTT 使用固定衰减 λ 不同，NDOT 使用动态的 e 来更新时间梯度 $\hat{a}$ ，这使得历史建模更精确，在复杂数据集上取得了更高的准确率。”

VI. REPRESENT WORK—S-TLLR: STDP-INSPIRED TEMPORAL LOCAL LEARNING RULE

S-TLLR: STDP-inspired Temporal Local Learning (2025)

Key Innovation: Incorporates both *causal* and *non-causal* (STDP) spike-timing relations.

Generalized STDP Eligibility Trace:

$$e_{ij}[t] = \alpha_{pre} \Psi(u_i[t]) \cdot \text{tr}(x_j)[t] + \alpha_{post} x_j[t] \cdot (\text{tr}(\Psi(u_i[t])) - \Psi(u_i[t])) \quad (10)$$

- **Causal term** (α_{pre}): Post fires *after* pre.
- **Non-causal term** (α_{post}): Post fires *before* pre.

Task-dependent Preference: Advantage: Memory $\mathcal{O}(n)$; leverages relations overlooked by BPTT.

Dataset	Optimal α_{post}	Temporal Nature
DVS Gesture	-1 (Causality)	Spatial-dominant
SHD	+1 (Synchrony)	Temporal-dominant

讲稿对应本页PPT:

”S-TLLR 由 Apolinario 和 Roy 于 2025 年提出，核心创新是引入了 STDP 中的非因果项。其资格迹包含两项：因果项（前发放后， α_{pre} ）和非因果项（后发放前， α_{post} ）。实验表明，在空间主导的视觉任务中，负的 α_{post} （强调因果）效果更好；而在时间主导的音频任务中，正的 α_{post} （强调同步）效果更好。这揭示了 BPTT 忽略的非因果关系在特定任务中的重要性。”

VII. REPRESENT WORK—TESS: TEMPORALLY AND SPATIALLY LOCAL LEARNING RULE

TESS: Temporally & Spatially Local Learning (2025)

Key Innovation: Achieves *full locality* (time & space) via local learning signals.

Time Locality (Eligibility Traces):

$$e_{pre}^l[t] = \alpha_{pre} \Psi(u^l[t]) \otimes q^l[t], \quad e_{post}^l[t] = \alpha_{post} h^l[t] \otimes o^{l-1}[t] \quad (11)$$

Space Locality (Local Signal Generator):

$$m^l[t] = B^{l\top} (f(B^l o^l[t]) - y^*) \quad (12)$$

- B^l : Fixed quasi-orthogonal binary matrix (square waves).
- $f(\cdot)$: Softmax (classification) or Identity.
- No backpropagation across layers: $\mathcal{O}(Cn)$ vs $\mathcal{O}(n^2)$.

Performance: Matches BPTT on CIFAR10/DVS Gesture with $205\times$ fewer MACs.

讲稿对应本页PPT:

”TESS 由 Apolinario 等人于 2025 年提出，实现了时间和空间的完全局部学习。时间局部性继承自 S-TLLR 的因果和非因果迹。空间局部性是其最大突破：使用固定的准正交二进制矩阵 B 将本层输出投影到任务子空间，与标签比较后投影回本地，生成学习信号 m ，完全不需要跨层反向传播。计算复杂度从 $\mathcal{O}(n^2)$ 降至 $\mathcal{O}(Cn)$ ，在保持 BPTT 精度的同时，MAC 操作减少了 200 倍以上。

VIII. DISCUSSION QUESTIONS

Discussion Questions

- 1) **Exact vs. Approximate:** RTRL gives exact gradients ($\mathcal{O}(n^3)$). OTTT/TESS approximate for efficiency. When would you sacrifice memory for exactness, and vice versa?
- 2) **Biological Plausibility:** E-prop uses “slow variables” (ALIF). S-TLLR uses “non-causal STDP”. How do these bio-mechanisms offer computational advantages over standard BPTT?
- 3) **Future of Local Learning:** TESS eliminates global backpropagation via random projections. Can fully local rules eventually surpass BPTT in generalization, or are they inherently limited by lack of global coordination?

讲稿对应本页PPT:

”讨论环节包含三个开放性问题。第一，精确与近似的权衡：何时该选择 RTRL 的精确梯度，何时该选择 OTTT 的效率？第二，生物合理性：E-prop 的慢变量和 S-TLLR 的非因果 STDP 如何提供超越标准 BPTT 的计算优势？第三，局部学习的未来：TESS 这类完全局部算法能否在泛化性上最终超越 BPTT，还是受限于缺乏全局协调？”

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